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Droplet energy harvesting system based on MXene/SiO_2 modified triboelectric nanogenerators

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ABSTRACT

Raindrops are an abundant micromechanical renewable energy source in nature, and raindrop energy harvesting has always been a research hotspot. Traditional hydropower generation is not suitable for low-frequency energy harvesting, and it is difficult to realize distributed energy harvesting. The development of triboelectric nanogenerator (TENG) is a promising strategy for converting raindrop energy in the environment into electrical energy. Polydimethylsiloxane (PDMS) composite has received much attention due to its excellent chemical stability and insulating properties for TENG. However, the deficiency of the low output performance and insufficient hydrophobicity of PDMS limits the application of PDMS in droplet energy capture. Modification of PDMS can improve the droplet energy collection capability of the liquid–solid contact triboelectric (LS-TENG). In this work, a new approach to modifying polydimethylsiloxane (PDMS) as a hydrophobic negative triboelectric material using a high dielectric constant MXene and a hydrophobic nano-SiO_2 was reported. The as-prepared composites' permittivity and triboelectric performance were remarkably promoted as compared to composites without interfacial modification. In addition, the use of composites for TENG assembly improved the open-circuit voltage and short-circuit current by a factor of 47.5 and 25.21, respectively, compared to pure PDMS. Application of the composite to the assembled TENG was employed as energy harvester to light up LEDs and charge a capacitor. In addition, a cathodic protection system was designed to demonstrate its potential application in the field of electrochemistry. This provides guidance for the practical implementation of TENGs in the field of electrochemical cathodic protection.

1. Introduction

The traditional electrochemical metal protection method is plagued by numerous problems, including significant environmental pollution and excessive waste of resources. Therefore, the search for a novel energy source to replace the conventional impressed current cathodic protection power supply has become a focal point of research [1,2]. Currently, various forms of renewable energy have been developed and utilized, including solar power, ocean energy; and wind power. Expensive equipment, complex processes, and environmental selectivity [3,4] have always been limiting factors in the development of green energy. Raindrops have both electrostatic and mechanical energy [5,6]. The efficient and easy development and utilization of raindrops is an inevitable trend in energy conversion, starting from the new modern energy system [7].

In 2012, Wang et al. [8–10] proposed and developed the TENG. TENG is a device based on triboelectrification and electrostatic induction [5,8,11–13], that realises the collection of randomly distributed, small low-frequency energy and converts it into electrostatic energy. TENG [14,15] have advantage of flexible structure, low cost, and versatile applicability. TENG has become a popular research area for scientists around the world. Recently, Kim et al [16 17] reviewed in detail recent research on ocean energy harvesting for hybrid systems and realized fast response, high sensitivity and wide range monitoring of humidity sensing using TENG doped with biocompatible polydopamine material. The potential of LS-TENG in raindrop energy harvesting has

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been demonstrated in previous studies [18–20]. The application of LS-TENG in water flow and raindrop energy collection can alleviate the energy problem. In the energy conversion process of LS-TENG, electricity can be generated by direct interaction between water and insulating objects without the need for complex conversion processes [21]. LS-TENG is considered to be a raindrop energy harvesting device, featuring prolonged lifespan, minimal material loss, environmental friendliness and high scalability [14,22]. The TENG technology exhibits the potential to harvest energy from small, dispersed raindrops [23,24]. The application of LS-TENG, combined with traditional cathodic protection, in the development and utilization of raindrop energy can realize a novel environmental-friendly, cost-effective cathodic protection system [25]. However, its practical application is limited by the low output current of LS-TENG and the power density, efficient collection of raindrops' energy remains a challenge, hindering its widespread use in daily life. Compared to the solid–solid TENG, the LS-TENG exhibits a more intricate manufacturing process and lower output performance. Rapid contact and separation of droplets on the surface of LS-TENG is necessary to mitigate electrostatic shielding phenomenon. To improve droplet propulsion and increase efficiency. For example Wang et al. [26] developed a fully enclosed LS-TENG featuring a U-shaped tube structure that converts low-frequency wave energy into electrical power. Nie et al. [27] fabricated elastic superhydrophobic cellulose paper by spray-coating nanofumed silica dispersed in a thermoplastic elastomer solution. Kim et al. effectively changed the surface potential of friction materials by studying the effect of argon ion injection on Kapton [28]. Despite the massive work devoted to fabricating TENGs for raindrops energy capture and developing applications of superhydrophobic properties, research on modifying PDMS for LS-TENG and exploring LS-TENG studies for raindrops energy harvesting is rare.

The electrical output performance of TENG is significantly influenced by friction material. Polydimethylsiloxane (PDMS) [29–33] is a commonly used insulating material for TENG, owing to its high electronegativity, flexibility, transparency, non-toxicity and cost-effectiveness. However, the use of PDMS as a triboelectric layer in LS-TENG presents certain challenges, including inadequate hydrophobicity, low surface roughness and suboptimal output performance. Improving hydrophobicity, electronegativity, and dielectric constant of insulating materials through surface modification [34,35] and composite material [36] is the primary method for improving the electrical output performance of materials.

In this study, we used ordered mesoporous silica nanoparticles with large specific surface area as effective charge storage sites in PDMS to enhance the charge storage capacity within the friction material [37,38]. The excellent dielectric properties and electronegativity of MXene were used to enhance the surface potential and charge density of the composites to create a hydrophobic negative triboelectric material. The modified PDMS was used to fabricate TENG. A single water droplet sliding over the surface of modified PDMS can generate an astonishing output current of 5.8 μA and an output voltage reaching up to 152 V. Compared to PDMS based TENG, the open voltage and shorted current of modified PDMS were improved by 47.5 times and 25.21 times, respectively. The addition of SiO_2 increases the surface roughness and hydrophobicity of PDMS. Droplets can quickly slide over the surface of modified PDMS. Water adsorption on the surface of the friction layer in environments with high relative humidity will dissipate the resultant charge, which is detrimental to the conversion of TENG energy. The enhancement of the hydrophobic surface ensures that the TENG stabilizes its performance in high humidity conditions. MXene improves dielectric properties and enhances charge accumulation in modified PDMS. The modified PDMS exhibits increased charge storage capacity and improved raindrop energy harvesting performance. In addition, the TENG cathodic protection system was developed to demonstrate its potential application in the field of electrochemistry. In the process of protecting 304 stainless steel (304SS), a corrosion potential drop of 200 m V was maintained, demonstrating the potential for green energy harvesting and self-powered sensing. This provides guidance for the practical application of TENG in the field of electrochemical cathodic protection.

2. Experiment

2.1. Materials and reagents

SYLGARD 184 silicone elastomer (polydimethylsiloxane, PDMS) was obtained by Matsui Trading Co., LTD. (Tianjin, China). NaCl and 95 % alcohol solution were obtained from Aladdin. 1H, 1H, 2H, 2H-perfluorodecyltriethoxysilane (FAS, $C_{16}H_{19}F_{17}O_3Si$) and gas phase SiO_2 were obtained from the Macklin reagent net. MXene (0.3 ~ 10 μm, Ti_3C_2x) was purchased from Jilin 11 Technology Co., Ltd. (Changchun, China). The test water is obtained by the water purification system of Xiamen Water Purification Technology Co., Ltd.

2.2. Preparation of the hydrophobic SiO_2

1H, 1H, 2H, 2H-perfluorodecyltriethoxysilane and alcohol were mixed in a 1:99 ratio into a fluorinated solution. Then, 0.2 g SiO_2 (30 nm ~ 50 nm) powder was added to the fluorinated solution and stirred at room temperature for 24 h. Finally, the fluorinated SiO_2 was dried in a vacuum drying chamber for 6 h to obtain the hydrophobic SiO_2 (F-SiO_2).

2.3. Preparation of the composite dielectric layer

SYLGARD 184 silicone elastomer was consisted of prepolymer A and curing agent B. Solutions A and B were weighed with a mass ratio of 10:1 as PDMS prepolymer. The dried F-SiO_2 is ground using a mortar. The treated F-SiO_2 (doping ratio of 3.21 %) was added to the prepolymer PDMS to obtain the mixture 1. Mixture 1 was ultrasonic for 30 min and stirred every 10 min in order to fully disperse F-SiO_2. Then, the MXene powder (0.321 % doping ratio) was added to the mixture 1 as functional transformation and sonicated for 15 min to obtain the mixture 2. The mixture 2 was dried in a 60 ℃ vacuum oven chamber for 2 h. After preliminarily curing and vacuum oven drying, the MXene/SiO_2@PDMS was ready, as shown in Fig. 1.Fig. 2.

To investigate the effect of the doped material on the electrical output performance of the composite dielectric layer. The MXene composite dielectric layer and the SiO_2 composite dielectric layer were prepared. Briefly, MXene (F-SiO_2) is doped into the PDMS and then cured to form MXene@ PDMS (F-SiO_2@ PDMS).

2.4. Preparation of MS@TENG

Fig. 2a-b show the schematic diagram of the MXene/SiO_2@PDMS-TENG (MS-TENG) device with a multi-layer composites structure. The effective contact area of the MS-TENG in the two-electrode mode was 96 cm^2. A commercially available copper tape (1 mm, 15 mm, 60 mm) was attached to a hard PET sheet (1 mm, 80 mm, 120 mm) as an induction electrode. A total; of four copper tapes were used. A two-electrode triboelectric nanogenerator is formed by connecting the first and third electrode as the positive pole of the TENG. The second electrode is used as the negative electrode by connecting the fourth electrode. The MXene/SiO_2@PDMS prepolymer was deposited on the rigid PET sheet at 60 °C. A 60 μm MXene/SiO_2@PDMS film was prepared by a blade coater. Finally, the composite film was cured on PET film to obtain MXene/SiO_2@PDMS-TENG (MS@PDMS-TENG). The PDMS-TENG (MXene-TENG、F-SiO_2-TENG) was prepared by the same procedure.

2.5. Materials characterization and electrical measurements

The micromorphology and element distribution of PDMS, MXene/SiO_2@PDMS, SiO_2@PDMS and MXene@PDMS were investigated by scanning electron microscope (SEM; HITACHI Regulus 8100, Japan),

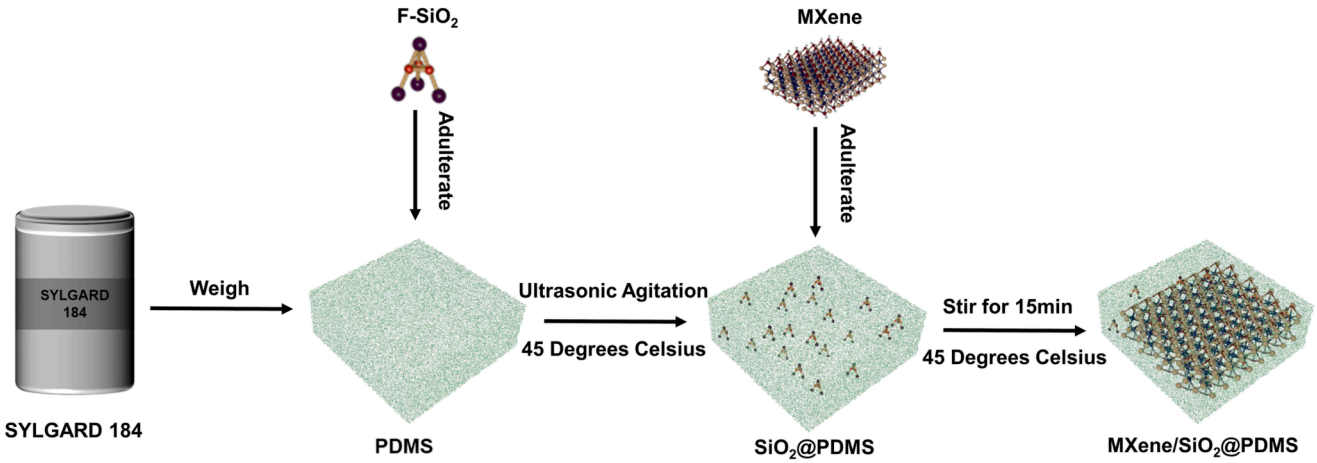


Fig. 1. Production flow chart of the MXene/SiO_2@PDMS.

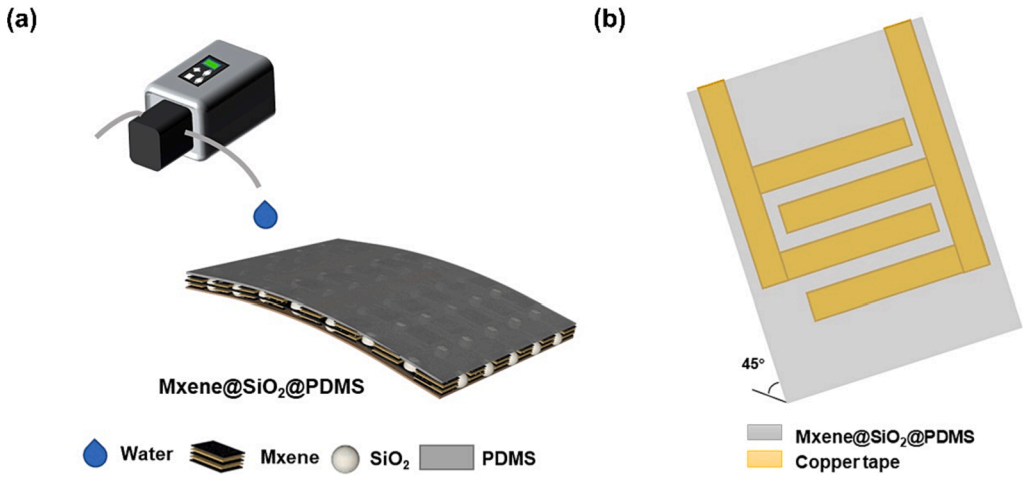


Fig. 2. a) structure of the mxene/sio$_2$@PDMS. b) Macroscopic device for MXene/SiO_2@PDMS at 45°.

and an energy-dispersive X-ray spectroscopy (EDS) system. The chemical compositions and elemental valence states of the samples were investigated using X-ray photoelectron spectroscopy (XPS; Thermo Escalab 250Xi, USA). 3D images of the samples were obtained using laser scanning confocal microscopy (LSCM; LEXT OLS5000, Japan).

Broadband dielectric spectroscopy (Novocontrol Concept 40, Germany) was used to investigate the dielectric characteristics of the samples in the frequency range of 10 Hz to 106 H_Z at room temperature. The

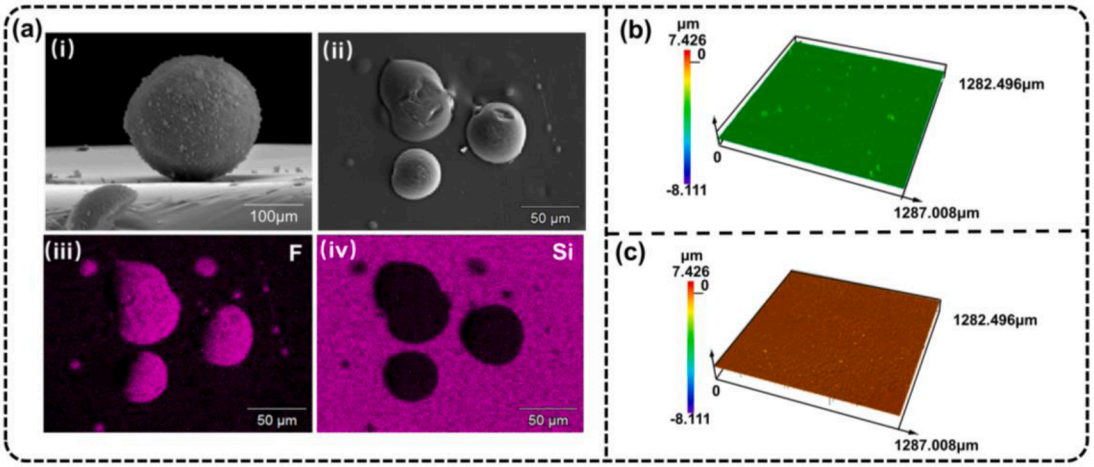


Fig. 3. a) sem and eds image of the mxene/sio$_2$@PDMS. i) Forms of SiO_2 on the surface of the triboelectric layer. ii) The SiO_2 embedded in the surface of the triboelectric layer. iii) Detection of Element F by EDS. iv) Detection of the presence of element Si by EDS. b) 3D morphology of pure PDMS. c) 3D morphology of the MXene/SiO_2@PDMS.

short-circuit current (Isc) and charge density of TENG were recorded using an electrometer K6514 (Keithley6514, Tectronix, USA). The open circuit voltage (Voc) is measured by an oscilloscope with a 100 MΩ high impedance probe (DS12022-E, RIGOL, China). Water droplets are produced by a Rong Bai constant flow pump (BT100 + YZ2515).

3. Results and discussion

3.1. Performance of MS@PDMS

As shown in Figure Fig. 3a (i)-(ii), the presence of F-SiO_2 on the PDMS surface is confirmed. The presence of hydrophobic F-SiO_2 increases the surface roughness and hydrophobicity of PDMS. Elemental mapping of SiO_2 on PDMS. The surface of SiO_2 is completely covered by F-elements, indicating that SiO_2 is completely fluorinated (Fig. 3a (iii)-(iv)). Fig.S2 shows the SEM of the surface and cross sections of the MXene/SiO_2@PDMS composite dielectric layer, which shows that MXene and SiO_2 are dispersed on the surface and inside of PDMS. A detailed analysis is given in Supplementary Note 1.

Fig. 3b-c show the 3D morphology of the dielectric layers of pure PDMS and MXene/SiO_2@PDMS. Pure PDMS has a smooth surface with no protrusions. The MXene/SiO_2@PDMS surfaces show different sizes of protrusions. These protrusions can cause an increase in the surface roughness of the MXene/SiO_2@PDMS. The TD-K401 microscopy images again confirmed the presence of the MXene/SiO_2@PDMS surface protrusion (Fig.S3). 3D images of other composite dielectric layers are shown in Fig.S4. The results show that the presence of MXene and FSiO_2 on PDMS increases the surface roughness. The increase of the surface roughness is favorable to increase the hydrophobicity of the MXene/SiO_2@PDMS composite films [39,40]. The roughness values of the composite film are shown in Fig.S5. The surface roughness of the MXene/SiO_2@PDMS films increased by a factor of 11.44 in the root mean square height, Sq, and a factor of 5.61 in the arithmetic square root height, Sa, compared to the surface roughness of the PDMS. The results show that the average surface roughness (Sa) and the standard surface roughness (Sq) of the composite membrane modified by MXene and SiO_2 are much higher than the pure PDMS. We observed the cross-section of the composite film by SEM and Mapping. The results confirm the reliability of MXene and SiO_2 uniformly dispersed in the composite film (Fig.S6). We specifically investigated this in Note 2.

To further investigate MXene/SiO_2@PDMS, FTIR and XPS were used to analyze the components. Fig. 4a shows the FTIR spectra of several friction materials. The surface of the MXene/SiO_2@PDMS composite membrane did not exhibit any new characteristic absorption bands. The results show that MXene and SiO_2 bind to PDMS in the form of a physical blending without strong chemical interactions [41–44]. Fig. 4b shows the XPS analysis of MXene/SiO_2@PDMS. The spectrum of MXene/SiO_2@PDMS contains six major elements (C, O, N, F, Si, and Ti). Only a single peak of the Ti-C bond was observed at 472 eV, which is due to partial breaking of the chemical bond in MXene due to ultrasonic agitation [42,45,46]. The Si_2p spectrum shows two distinct peaks at 102.48 eV and 101.81 eV, corresponding to the $Si_2p_{1/2}$ and $Si_2p_{3/2}$ states, respectively. The XPS spectra of Cl_s show two peaks. The peaks detected at 284.61 and 286.09 eV correspond to the C-Si and O-C@O functional groups, respectively. The centres of O1s are 532.34 eV and 532.92 eV, corresponding to O-Si and O-C functional groups, respectively. The peak of F1s at 689.08 is attributed to C-Si-F. The results demonstrate the successful binding of MXene and SiO_2 to the PDMS. In Fig.S7, the XRD composition of MXene/SiO_2@PDMS was tested and analyzed.

3.2. Structure design and working principle of the MS-TENG

Fig. 5a shows the working mechanism of triboelectrification and electrostatic induction of MS-TENG. A water droplet is used as an illustrative example to demonstrate the four stages of the working process of MS-TENG. The droplet was positively charged when it contacted from the MXene/SiO_2@PDMS surface. When the droplet touches the negative triboelectric material, the effect of the contact charge leads to the positive and negative polarity in the water droplet and the triboelectric layer [47]. Negative charges accumulate on the underside of the triboelectric layer, namely, the first copper electrode, negative charges accumulate and effectively shielding the positive charge of the droplet. The other three electrodes in the MS-TENG are positively charged to neutralize the negative charge in the triboelectric layer. As the droplet begins to slide along the surface of the triboelectric layer, the accumulated negative charge level begins to flow along the copper electrode, generating an electric current. When the droplet reaches the top of the second copper electrode, the negative charge accumulates, effectively shielding the droplet for positive charge. As the droplet continues to flow over the surface of the triboelectric layer, the accumulated negative charge flows in the opposite direction along the copper electrode. Similarly, when the droplet reaches between two neighboring copper electrodes, the accumulated charge moves back and forth, generating a large amount of alternating current.

Fig. 5b shows the mechanistic diagram of the enhanced hydrophobic performance of the MX-TENG liquid–solid interface. The microstructural protrusions generated on the modified PDMS surface increase the PDMS surface roughness. This increase in roughness results in a larger water contact angle. Water droplets fall on the surface of the composite friction layer, they cannot penetrate all the way into the pores under the capillary force. Therefore, an air cushion is formed between the water droplets and the rough structure by trapping air, which makes the solid–liquid interface change to a mixed solid–liquid and gas–liquid

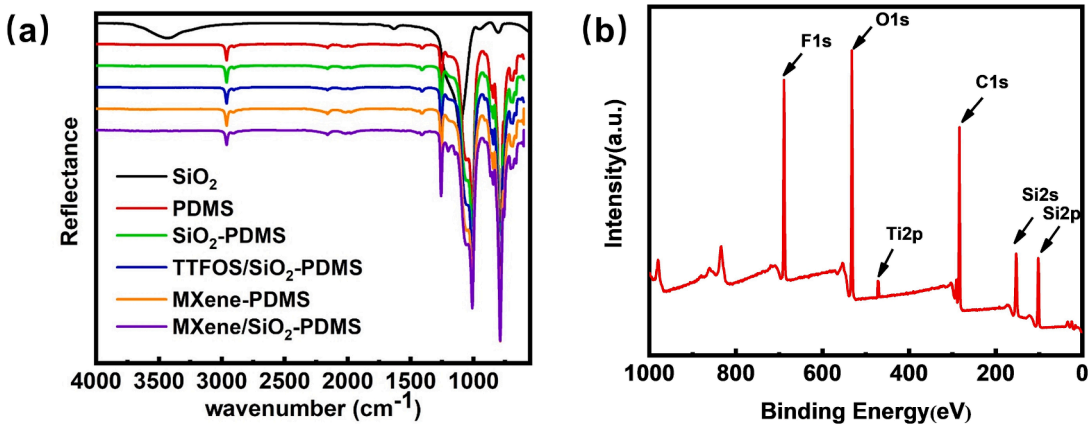


Fig. 4. a) infrared spectra of pdms, sio$_2$, SiO_2@PDMS, MXene@PDMS, MXene/SiO_2@PDMS. b) XPS spectrum of MXene/SiO_2@PDMS.

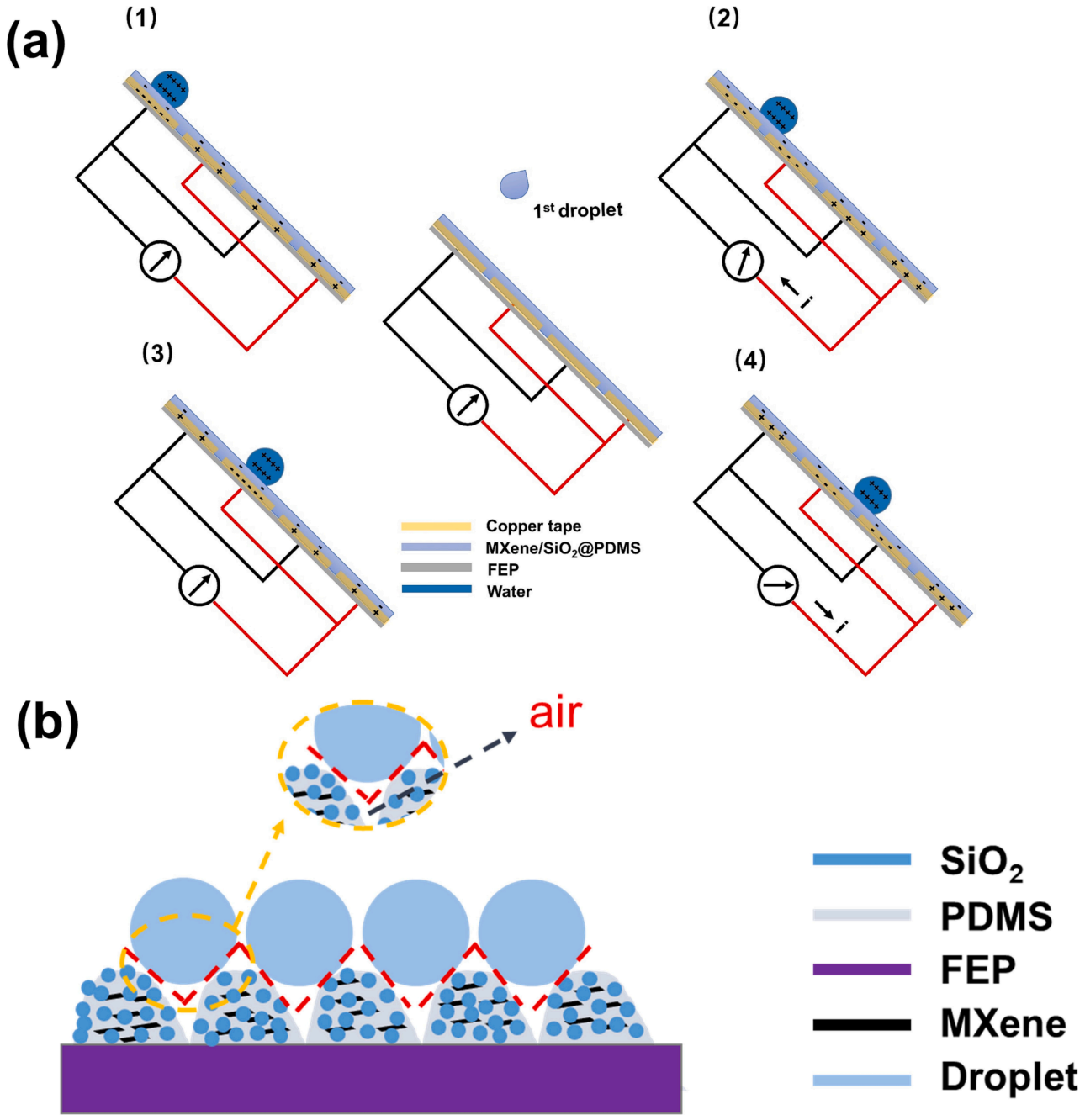


Fig. 5. a) the operation principle of ls-teng. b) schematic diagram of mxene/sio$_2$@PDMS hydrophobicity enhancement.

interface. The presence of the air–liquid interface increases the water contact angle of the film. This increase in performance accelerates the rapid sliding of the droplets on the surface of the friction layer and reduces the charge shielding of the droplets on the surface of the insulating layer. Meanwhile, SiO_2 can enhance the charge storage capacity of PDMS due to its large specific surface area. In this process, the combination of MXene material with high conductivity [48] and pseudo-capacitance with PDMS reduces the surface charge shielding and enhances the polarization of the polymer surface, which in turn increases the charge density on the PDMS surface. The improved performance of MXene/SiO_2@PDMS can be attributed to the combined effect of MXene and SiO_2. During the design process of LS-TENG, water droplets need to leave the surface as soon as possible after impacting the surface of the composite layer in order to ensure the efficient operation of the generator. Although superhydrophobic materials can realize rapid contact separation of droplets, they suffer from poor weathering resistance, fast aging and instability. Experimentally, we found that the dielectric layer prepared by doping some hydrophobicity can also achieve good results.

3.3. Output performance of MS@TENG

Fig. 6a-b shows the short circuit current and open circuit voltage of TENG composed of different modified triboelectric materials. The results show that the V_{oc} and I_{sc} of MS-TENG are 152 V and5.8μA. Compared with PDMS-TENG (V_{oc} = 3.2 V, I_{sc} = 0.23 μA) , the V_{oc} and I_{sc} are increased by 47.5 times and 25.21 times, respectively. The V_{oc} and I_{sc} of MXene-TENG were 26 V and 0.49 μA, respectively. The V_{oc} and I_{sc} of F-SiO_2-TENG were 28 V and 1.2 μA, respectively. The results show that MS-TENG has the optimal potential output performance. In Fig. S8, the results show that the peak asymmetry of the positive and negative

currents is determined by the time of separation of the droplet from the metal electrode. Electron affinity, surface hydrophobicity, and ionization energy are important parameters used to evaluate the performance characteristics of various triboelectric materials [49–51]. The improvement in the output performance of MS-TENG is mainly attributed to the improvement in contact angle and dielectric performance. In Fig. 6a, the wetting properties of four triboelectric materials were characterized. The results show that the contact Angle of PDMS is increased from 90° to 123° after modification. The increase in contact angle ensures that the droplet can quickly slide on the surface of MXene/SiO_2@PDMS, reducing the charge shielding, and improves the electrical performance output [52]. The increase in contact angle is attributed to the deposition of F-SiO_2 on the PDMS surface.

To verify the mechanism of enhanced electrical output performance of TENG composed of MXene/SiO_2@PDMS composite film. Fig. 6d-f show the dielectric constants, dielectric losses and insulation properties of several composite films. The dielectric constant (εr) of MXene/SiO_2@PDMS shows a significant increase compared to that of unmodified pure PDMS [53,54]. At a frequency of 1000 Hz, the F-SiO_2@ PDMS, MXene@PDMS, and MXene/SiO_2@PDMS were measured to be 3.88, 4.212 and 4.566. Compared to PDMS with a dielectric constant of 2.83, the dielectric constants of SiO_2@PDMS, MXene@PDMS and MXene/SiO_2@PDMS increased by 148.56 %, 137.12 % and 161.04 %, respectively. The dielectric loss of MXene/SiO_2@PDMS is extremely low (Fig. 6e). The dielectric constant of triboelectric materials plays an important role in increasing the charge density [55]. The charge density can be expressed by the formula:

$$\sigma' = \frac{\sigma_0 d_{gap}}{d_{gap} + d_{PDMS}/\varepsilon_{PDMS}} \quad (1)$$

σ^0, d_{gap}, d_{PDMS} and ε_{PDMS} are the triboelectric charge density of the equilibrium state, the gap distance, the thickness of the triboelectric material and the dielectric constant of the triboelectric material, respectively. From Equation (1), σ′increases with the dielectric constant. Therefore, high dielectric constant and low dielectric loss are the key factors for achieving high performance output of MS-TENG. Moreover, the Fig. 6f results show that the MXene/SiO_2@PDMS composite dielectric film has enhanced conductivity at high frequencies, and this improved performance can be attributed to the high conductivity and work function of MXene. The enhancement of conductivity facilitates the prompt transfer of electrons, thereby preventing excessive charge accumulation on the surface of dielectric layer and impeding performance output.

The addition of MXene increases the charge density on the surface of the composite film and increases the dielectric constant [56]. Excessive doping causes stacking between the MXene, which generates leakage currents. This situation leads to a decrease in the charge density on the surface of the composite film, resulting in a decrease in TENG performance. Excessive doping of MXene reduces the electrical performance of the modified membrane, and Fig. 6g-h show the effect of MXene particle doping on the performance of MS-TENG. The open circuit voltage and short circuit current of MS-TENG exhibit an initial increase followed by

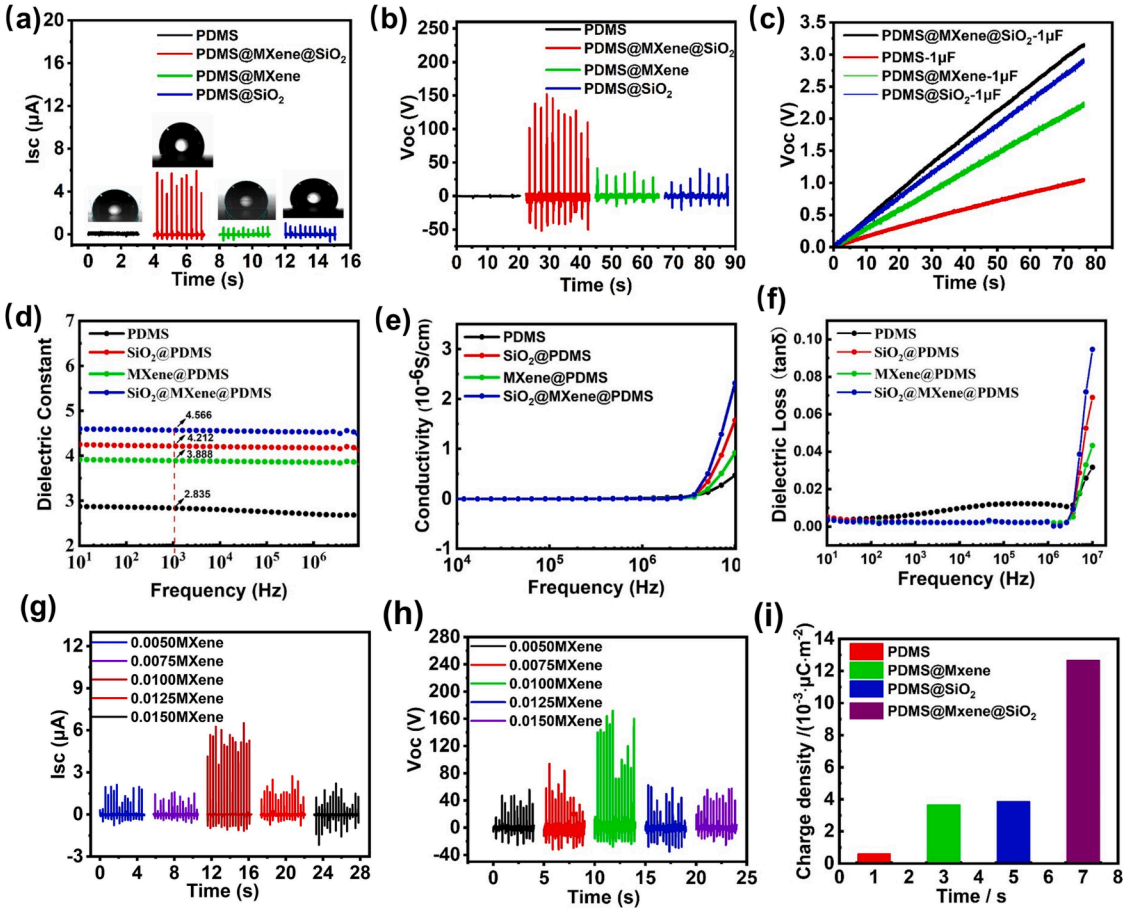


Fig. 6. Output characterization of MS-TENG. a) Output characterization current of the TENG with different negative triboelectric materials. b) Open circuit voltage to TENG by different negative triboelectric materials. c) The charge characteristic curves of TENG with different negative triboelectric materials. d) Dielectric constant of a composite dielectric layer. e) Dielectric loss in composite dielectric layers. f) Insulating properties of composite dielectric layers. g) Output current of TENG at different MXene doping concentrations. h) Open-circuit voltage of the base TENG with different MXene doping concentrations. i) Integral charge of various modified films.

a subsequent decrease as the concentration of MXene doping is increased. MXene doping concentration at 0.6 % leads to highest output performance for MS-TENG. The role of F-SiO_2 is to regulate the hydrophobicity of the surface of the composite dielectric layer [57]. We added a certain amount of F-SiO_2 to PDMS to increase the chance of air pockets forming between water droplets and PDMS, thus increasing the hydrophobicity of the PDMS surface. In Fig.S9 a-b, the results show that MS-TENG has the highest output performance when the SiO_2 doping concentration is at 3.21 %. Fig. 5c shows the charging capacity of the TENG composed of different friction materials within 75 s (rectified before charging). The results demonstrate a higher charging performance of MS-TENG. Fig.S10 shows the influence of different doping concentrations on the charging performance of MS@TENG, and the evaluation of different capacitor charging capacities. The results are consistent with the Fig. 6a. Fig. 6i shows the maximum output charge of a drop of water on the TENG under the optimal test conditions. The results show that a drop of water can transfer a larger amount of charge on the surface of MS-TENG. Fig. S11 shows the integrated charge of MS-TENG at different experimental angles, with the largest transfer charge observed at 45°. In conclusion, the results suggest that MS-TENG exhibits superior electrochemical performance compared to PDMS-TENG. It's worth noting that the output performance of the MS-TENG is comparable to, if not better than, that of the LS-TENG before it (Table S1). Video 1 shows the process of harvesting droplet energy.

3.4. Electrical output characteristics of the MS-TENG

Fig. 7a, b shows the changes in output current and voltage as the peristaltic pump speed is increased from 10 rpm to 50 rpm with MS-TENG at 45°. The results show that there is no change in I_{sc} and V_{oc} during the speed change. This phenomenon is attributed to the absence of the superhydrophobic MXene/SiO_2@PDMS material. The number of droplet impacts on the friction material surface per unit time is controlled by the droplet velocity. Higher rotational speeds would result in more droplets hitting the MXene/SiO_2@PDMS surface per unit time, resulting in more charge. Video 2 shows a comparison of the sliding angle of a droplet on a film before and after modification. Fig. 7c-d shows the change in output performance with different test angles at 50 rpm. The results show that the output performance is highest at the 45° test angle (I_{sc} = 5.8 μ A, V_{oc} = 152 V). When the test angle exceeds 45°, the output performance continuously with increasing angle, reaching 2.7 μA and 94 V at 60° , 2.1 μA and 76 V at 90°. Fig. 7e shows the change of current and voltage of MXene/SiO_2@PDMS with different external resistances at 50 rpm, 45° test angle. The voltage amplitude increases proportionally with the external resistance, while the current is inversely proportional. Fig. 7f shows the MXene/SiO_2@PDMS with a peak output power of 1.01 × 10^{-4} W at a resistance of 40 MΩ.

3.5. Application of MS@TENG

Fig.S12 shows the stability test for MS-TENG. The results show that MS-TENG still has excellent output stability after a long period of operation. LEDs light test was performed to validate the power capacity of MS-TENG. MS-TENG can instantly light up 21 commercial green LEDs when water droplets are at 50 rpm (Fig. 8b). A self-powered electrochemical cathodic protection system based on MS-TENG was designed. This system demonstrates the substitutability of MS-TENG for the traditional cathodic protection external power supply. As shown in Fig. 8a, after the output current is rectified, the positive electrode of MS-TENG is connected to a platinum electrode, and the negative electrode is connected to 304S. A saturated calomel electrode (Thunder electromagnetic) was used as the reference electrode. The open circuit potential of the protected metal is an important parameter to evaluate the effect of cathodic protection [58]. Fig. 8c shows the monitoring of the long-term potential difference between the protected metal and the reference electrode in electrolyte solution of 3.5 % NaCl. The results show that the Open circuit potential measurements (OCP) value of 304SS with external MS-TENG decreases from −180 mV to −380 mV, and the potential decreases by 200 mV. The interface charge transfer behavior of 304SS in solution was explored by electrochemical impedance spectroscopy (EIS) to further verify the reliability of MS-TENG for cathodic protection. Fig. 8d shows Nyquist diagrams with and without external MS@TENG cathodic protection and corresponding equivalent circuits in 3.5 % NaCl solution. R_s is the solution resistance, and Q_c and R_f are the capacitance and resistance of the 304SS surface membrane. In addition, R_{ct} and Q_{dl} are charge transfer resistance and double layer capacitor respectively. The cathodic protection effect of MS@TENG was evaluated by R_{ct} value. The R_{ct} value of 304 stainless steel with MS-TENG (2.689E5/Ωcm^2) is lower than that of 304SS without MS-TENG. This shows that when 304SS is coupled to MS-TENG, the electrons provided by MS-TENG are transferred to the 304SS surface, resulting in an electron excess on the 304SS surface [1,59]. TENG-connected 304SS

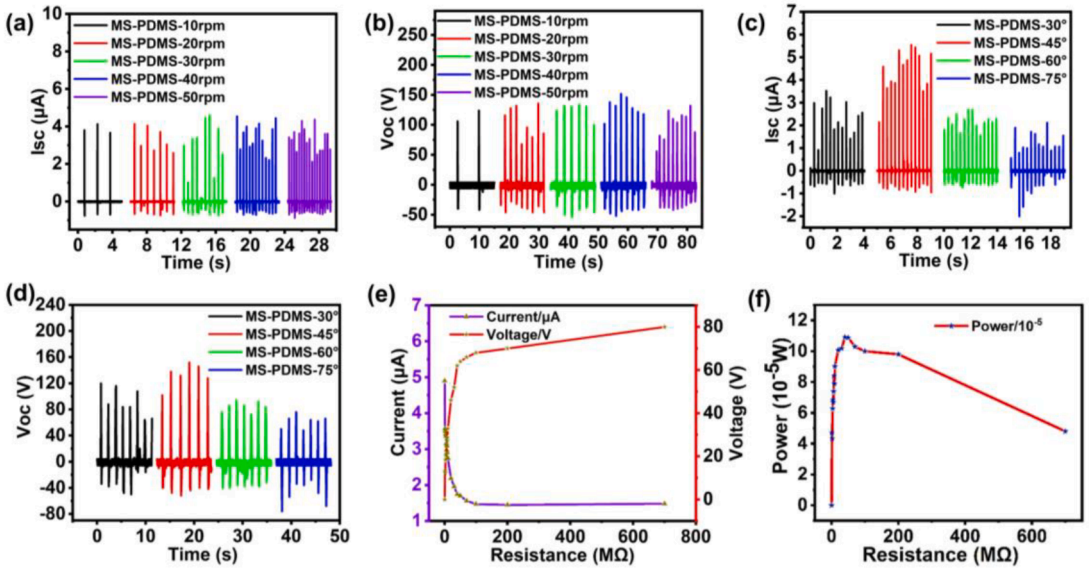


Fig. 7. a) i_{sc} of MS-TENG at different flow rates. b) V_{oc} of MS-TENG at different flow rates. c) Output current of MS-TENG at different angles. d) V_{oc} of MS@TENG at different angles. e) MS@TENG Current Voltage vs. Resistance. f) Output power of MS@TENG at different external circuit load resistances.

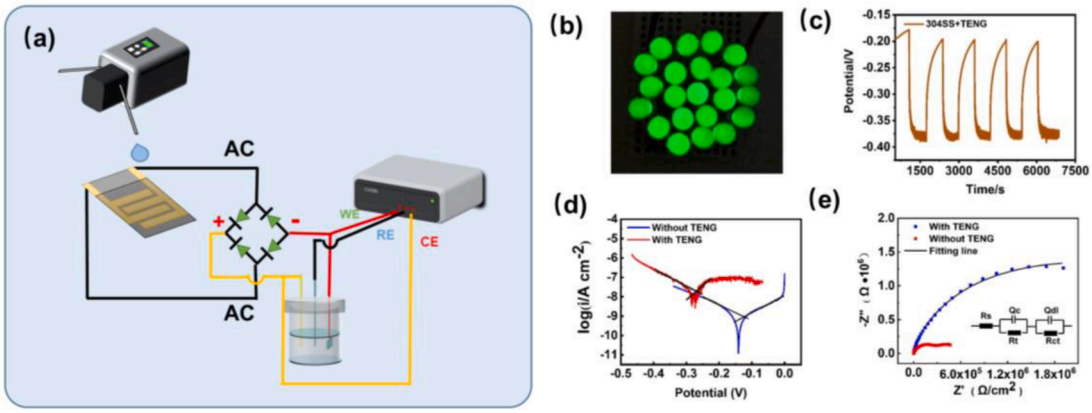


Fig. 8. a) rectifier ms-teng cathodic protection actual diagram. b) ms-teng power lighting led experiment. c) ocp value versus time curve for 304ss cathodic protection under 50 rpm output. d) Nyquist plot of 304SS with or without MS-TENG plus corresponding equivalent circuit. e) Polarization curves of 304SS with or without MS-TENG.

showed an increase in corrosion current density compared to TENG-connected 304SS without TENG. This finding is attributed to the electrons generated by the solid–liquid TENG array being injected into the 304SS surface to accelerate the electrochemical reaction. The above results are in agreement with the former OCP results shown in Fig. 8 c. R_{ct} also represents the charge transfer resistance and the electron transfer rate [14,6054,55]. Transfer of electrons from the external output of TENG to the protected metal surface results in reduced R_{ct} values. Fig. 8e result shows the increased slope of the 304SS anode (β_a) with MS-TENG and the decreased cathode slope (β_c). This indicates that the precipitation rate of Fe^{2+} decreases and the anode reaction accelerates. The corrosion reaction of stainless steel in high concentration solution can be simply expressed as:

$$Fe \rightarrow Fe^{2+} + 2e^- \quad (2)$$

$$O_2 + 4H_2O + 4e^- \rightarrow 4OH^- \quad (3)$$

$$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2 \quad (4)$$

$$4Fe(OH)_2 + O_2 + 2H_2O \rightarrow 4Fe(OH)_3 \quad (5)$$

The anode and cathodic reactions of the 304SS corrosion in the natural state correspond to steps (2) and (3), respectively. After 304SS of MS-TENG connection, the electrons generated by MS-TENG have an inhibitory effect on step (2), reducing the corrosion rate of 304SS. Step (3) explains the natural diffusion of oxygen to the 304SS surface and electron reaction, producing a large amount of OH–, which can enhance the rust resistance of 304SS to the corrosion medium. Therefore, external MS-TENG can have an effect of cathodic protection against 304SS. In conclusion, the results provide a reliable basis for MS-TENG for self-powered cathodic protection.

4. Conclusions

In this study, MXene/SiO_2@PDMS composites were successfully synthesized by doping nanoparticles into PDMS. The TENG based on MXene/SiO_2@PDMS demonstrated the potential of this novel material in various applications. The physically doped method forms a stable combination of MXene and SiO_2 with the PDMS. The addition of F-SiO_2 improves the surface roughness and hydrophobic properties of PDMS, which ensuring that the droplets can quickly slide on the MXene/SiO_2@PDMS surface, reducing the charge shielding of the surface. The addition of a small amount of MXene increases the electronegativity and conductivity of the membrane surface. The MXene powder is ultrasonically dispersed into the PDMS film to increase the electron storage depth and total charge density. Using MXene/SiO_2@PDMS to TENG, a single droplet can generate an output current of up to 5.8 μA and an output voltage of up to 152 V, demonstrating its potential for efficient energy harvesting from droplets. The potential of 304SS shifts to −380 mV after the MS-TENG device is applied to cathodic protection, and the potential reduction exceeds 200 mV, which is a significant protection effect. A designed cathodic protection system powered by MS-TENG demonstrates its self-powered cathodic protection capability.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2023.146832.

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